



JAIPUR NATIONAL UNIVERSITY

Semester/Year End (Main/Back) Examinations, May-2025

Name of School:	School of Engineering and Technology
Name of Programme / Course:	B.Tech. (Computer Science Engineering) / B.Tech. CSE(Hons.) in Artificial Intelligence and Machine Learning / B.Tech. CSE(Hons.) in Cyber Security
Semester/Year:	IV / II
Paper Code:	BTECSCO402T22
Title of Paper:	Object Oriented Programming using C++
Time: 3 Hrs.	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

This question paper consists of two sections, Section-A and Section-B. Section-A consists of seven questions and Section-B consists of six questions. Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question.

Question Number	SECTION-A	Marks Allotted
Q.1	Define a class Book with data members title, author, and price. Include a constructor to initialize the values and a member function display() to show the book details. Create an object and demonstrate its usage.	4
Q.2	What is the difference between shallow copy and deep copy in C++? Explain with an example using dynamic memory allocation.	4
Q.3	What is function overloading? Write a class Calculator that overloads the function add() to perform addition of: • Two integers • Two floats • Three integers	4

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| Q.4 | What is a copy constructor? How is it different from a default constructor? Write a class Student that demonstrates both default and copy constructors. | 4 |
| Q.5 | What will happen if no catch block matches the thrown exception? Explain with an example | 4 |
| Q.6 | Differentiate between ifstream, ofstream, and fstream with examples of how each one is used. | 4 |
| Q.7 | What is exception handling? Write a program in C++ that uses try, catch, and throw to handle division by zero. | 4 |

SECTION-B

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| Q.1 | Explain the concept of multiple catch blocks in C++ with an example. | 10 |
| Q.2 | Explain the role of constructors and destructors in C++. Write a program that demonstrates both parameterized and default constructors in a class called Student. Include the destructor to release any allocated memory. | 10 |
| Q.3 | Define an abstract class Shape with a pure virtual function area(). Derive two classes Circle and Rectangle and implement the area() function in both. Demonstrate polymorphism by calling area() using a base class pointer. | 10 |
| Q.4 | What is the Standard Template Library (STL) in C++? Explain its components with suitable examples for containers , algorithms , and iterators . | 10 |
| Q.5 | What is the role of try, catch, and throw in exception handling? Explain with a code snippet. | 10 |
| Q.6 | What is aggregation in object-oriented programming? How is it implemented in C++? Explain with an example where a Department class contains a Teacher object. | 10 |